

Finite Elements Analysis in Biomedical Engineering
Spring 2017, 376-1712-00

Assignment 1:
2D Truss Solver

Short report and simulation results

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Files:

The archive named **truss2D_Pepic.rar** (14 kB total) contains all of the MATLAB® scripts used for the assignment and should run on independently when extracted. List of files in the archive is the following:

1. **truss2Dcomplete.m** – main file from which functions are called and is to be executed first
2. **truss_example_assignment1_task[1 2].m** – contain input data for the solver (list of nodes, elements, loads and boundary conditions)
3. **plotInput.m** – function which plots the input data (non-deformed structure with node and element numbers and the loads)
4. **truss2D.m** – function which solves for nodal displacements and internal forces given the input data
5. **plotScale.m** – function which determines the necessary plot scale for the displacements (alternative to manual input, set to value which would show maximum displacement as 10% of the maximum length of the elements – similar is done for force vectors)
6. **plotDeformed.m** – function which plots the non-deformed and the deformed structure, with displacements scaled up to make them easily visible
7. **plotForces.m** – function which plots the (scaled) deformed structure and the values of internal forces in each element

Overall structure of the code:

% Loading of input data from the provided file(s)

```
truss_example_assignment1_task2;
```

% Plotting of the input data

```
plotInput(nodes, elements, load1, load2, BCs);
```

% Solving for the nodal displacements and internal forces in elements

```
[displacements,forcesInternal] = truss2D(nodes, elements, load1, load2, BCs);
```

%% Plotting the deformed and the non-deformed structure

```
plot_scale = plotScale(nodes,elements,displacements);
```

```
plotDeformed(nodes, elements, displacements, plot_scale);
```

%% Plotting of the deformed structure and the internal forces in elements

```
plotForces(nodes, elements, displacements, forcesInternal, plot_scale);
```

Notes on the FEM solver (truss2D function):

Input data has Young's modulus given in MPa, cross-section given in millimeters squared and loads given in Newtons – therefore the outputs of the solver are nodal displacements given in millimeters and internal forces in elements given in Newtons.

Since the input data was given with “linear” elements, the solver in the current form works for this type of elements only. The underlying equation used to obtain the solutions is the local stiffness equation (Hooke's law) of a single element:

$$\frac{AE}{L} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} d_1 \\ d_2 \end{pmatrix} = \frac{qL}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} P_1 \\ P_2 \end{pmatrix}$$

, which is valid in 1D (along the element) and where the variables in the equation are:

- A - cross section of the element (5th column of the elements matrix)
- E is the Young's modulus of the element (4th column of elements matrix)
- L is the length of the element (calculated from known coordinates of nodes defining the element)
- d_1 and d_2 are the local nodal displacements
- q is the distributed load acting along the element (e.g. weight of the element itself)
- P_1 and P_2 are the concentrated forces acting at the two nodes defining the elements

This equation needs to be “translated” into a 2D form, that is, the displacements and forces need to be separated into vertical and horizontal components in the given global coordinate system. It can be shown (lecture 2, slide 8 and 9) that in the global system the above equation ($\mathbf{k}_l \mathbf{d}_l = \mathbf{r}_l$) can be rewritten as:

$$(\mathbf{T}^T \mathbf{k}_l \mathbf{T}) \mathbf{d}_l = \mathbf{T}^T \mathbf{r}_l$$

$$\mathbf{K}_l \mathbf{d}_l = \mathbf{T}^T \mathbf{r}_l$$

, where T is the transformation matrix between global and local nodal degrees of freedom of an element:

$$\begin{pmatrix} d_1 \\ d_2 \end{pmatrix} = \begin{pmatrix} l_s & m_s & 0 & 0 \\ 0 & 0 & l_s & m_s \end{pmatrix} \begin{pmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{pmatrix}$$

, or shortly:

$$\mathbf{d}_l = \mathbf{T} \mathbf{d}$$

, and there $\mathbf{K}_l$ and $\mathbf{T}^T \mathbf{r}_l$ represent the local entries into the global stiffness matrix $\mathbf{K}$ and global force matrix $\mathbf{r}$, respectively.

Upon adding contributions of each elements in global stiffness and force matrices, boundary conditions (known nodal displacements) are taken into account by setting the line corresponding to that nodal displacement to zero, diagonal element to one, and the corresponding global force vector to the known nodal solution.

Upon solving the matrix equation $\mathbf{K} \mathbf{d} = \mathbf{r}$ solutions for nodal displacements (or NDoF in general) are obtained. From those the internal forces can be calculated as:

$$\mathbf{f}_l = \mathbf{k}_l (\mathbf{d}_2 - \mathbf{d}_1) = \mathbf{k}_l \mathbf{T}^T \mathbf{d}_{global, nodes\ 1\ and\ 2}$$

Results:

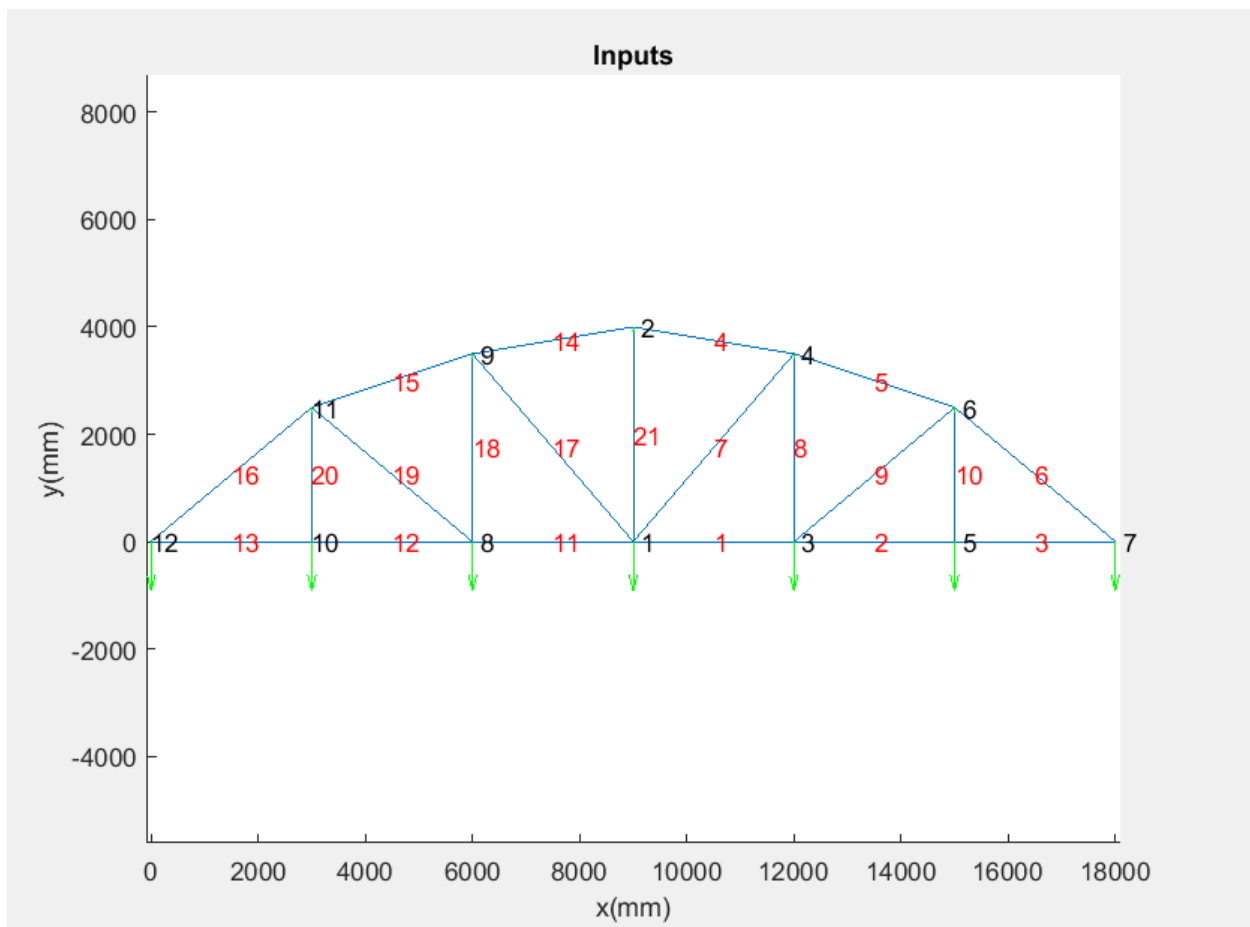


Figure 1: Plot of input data for task 2 (note the small green vectors in arc nodes)

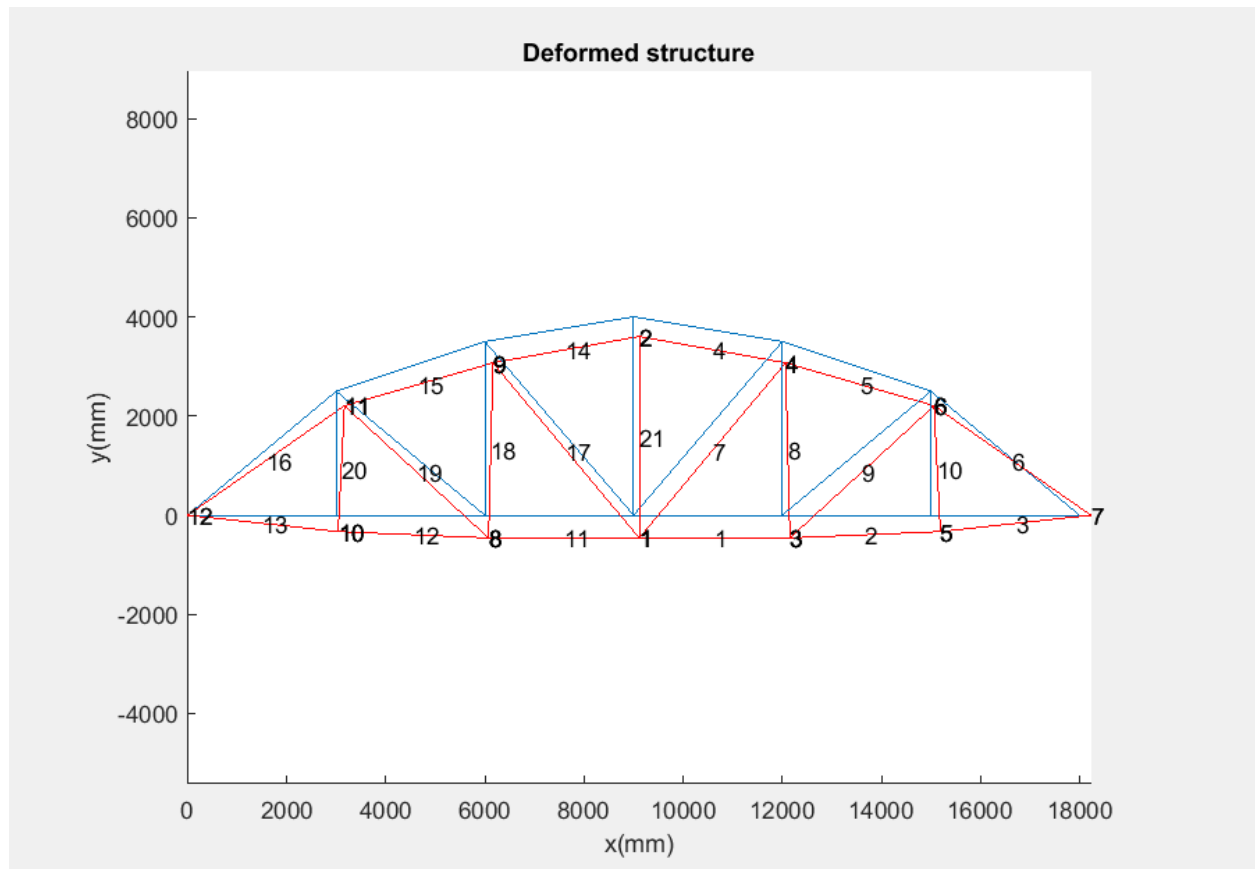


Figure 2: Plot of deformed structure for task 2. As expected, the bridge sags down due to its own weight and weight of the car crossing over it

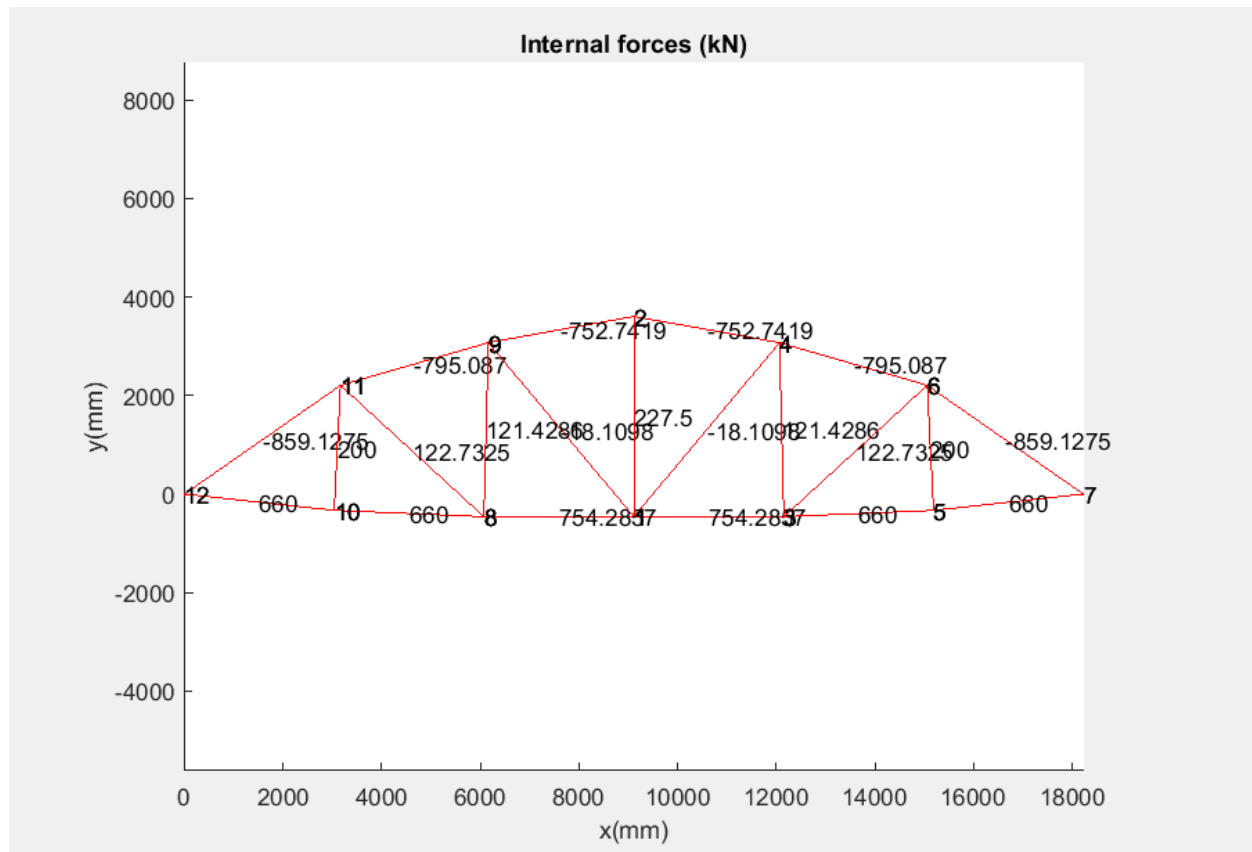


Figure 3: Plot of deformed structure and internal forces in the elements. As expected from forces in a single beam bending down, the bed segments are under tension since the bridge bends down while the arc segments are under compression. Forces in Internal segments are however not as intuitive to understand.

Literature:

1. Benedikt Helgason, Stephen J. Ferguson; Finite Elements Analysis in Biomedical Engineering course; Lecture 1, Lecture 2, Assignment 1; Spring 2017, ETH Zurich
2. MATLAB Documentation - MathWorks® Switzerland